



Executive Summary

This feasibility study addresses the lack of accessibility in e-learning platforms for visually impaired learners by proposing the development of Accendo, an Adaptive Learning Management System (LMS) powered by AI-based Natural Language Processing (NLP) and multi-modal haptic feedback. The system integrates speech recognition, text-to-speech, and tactile cues to provide intuitive, non-visual interaction, enabling learners to navigate and comprehend digital content independently. Its main objective is to bridge the accessibility gap in higher education by fostering inclusion and empowerment through technology.

The scope of the study covers the design, iterative development, and evaluation of the LMS using the Agile Development Methodology and guided by the ISO/IEC 25010 quality model. However, the project is delimited to visually impaired learners and constrained by resource availability, device compatibility, and academic timelines. Despite these risks and limitations, the system offers a credible and innovative solution to enhance accessibility, usability, and inclusivity in higher education e-learning environments.

CHAPTER 1 BUSINESS PROFILE ACCENDO

The name “Accendo” originates from the Latin word meaning *“to ignite,” “to illuminate,” or “to inspire.”* This name reflects the company’s purpose of bringing light and empowerment through accessible education.

Aligned with our vision to be a pioneer in inclusive digital education, Accendo symbolizes the spark of knowledge and innovation that illuminates learning opportunities for visually impaired learners. It represents our commitment to making accessibility and equity central to the future of e-learning systems.

In harmony with our mission, Accendo embodies our goal to design AI-powered adaptive learning systems that enable visually impaired students to learn independently and confidently. Through the integration of Natural Language Processing and multi-modal haptic feedback, Accendo seeks to ignite change within the education technology community—fostering fairness, inclusivity, and a brighter path toward accessible digital learning.

Nature of Business

Our company operates in the Educational Technology (EdTech) industry, focusing on the development of accessible and adaptive e-learning systems for visually impaired learners. The core business revolves around the design, manufacturing, and delivery of



digital solutions that combine Artificial Intelligence (AI), Natural Language Processing (NLP), and Multi-Modal Haptic Feedback technologies to make learning more inclusive and engaging.

The company integrates three major business areas:

1. Manufacturing – Development and integration of adaptive learning management software, AI-powered voice interaction tools, and haptic feedback systems that enhance accessibility for visually impaired learners.
2. Trading – Distribution and licensing of adaptive learning software packages, accessibility modules, and assistive devices to schools, universities, and government agencies.
3. Service – Providing technical support, user training, accessibility consulting, and continuous maintenance to ensure optimal system performance and compliance with international standards for inclusive education.

Through these combined areas, the company aims to promote digital inclusion and equal educational opportunities for learners with disabilities, aligning with global educational accessibility initiatives.

Company Vision and Mission Statements

VISION

To be recognized as a pioneer in inclusive digital education by creating opportunities for visually impaired learners through AI and haptic-based technologies, making accessibility and equity a core part of future e-learning systems.

MISSION

Accendo strives to make e-learning more inclusive by designing adaptive learning management systems that use AI-powered language processing and haptic feedback. Our goal is to help visually impaired learners study independently, support teachers with accessible tools, and encourage the education technology community to focus on accessibility and fairness in digital learning.

Company Objectives

In the short term, the company aims to develop and pilot an AI-powered Adaptive Learning Management System (ALMS) that integrates natural language processing and



multi-modal haptic feedback to enhance accessibility for visually impaired learners. Within the first two years, it seeks to establish partnerships with local educational institutions and accessibility organizations for testing and deployment. Alongside system development, the company will focus on building a strong brand identity as an innovator in accessible education technology in the Philippines and forming a multidisciplinary team skilled in artificial intelligence, user accessibility design, and software development.

For the long term, the company envisions expanding the implementation of the adaptive LMS across schools and universities nationwide, ensuring that visually impaired students can access high-quality, inclusive education. It plans to enhance the system to accommodate learners with other disabilities, introducing features that promote universal accessibility. The company also intends to launch a subscription-based platform for wider market access and global adoption, positioning itself as a leading provider of accessibility-driven educational technology in Southeast Asia. Continuous improvement will remain a core focus, guided by the ISO/IEC 25010 Quality Standard to ensure usability, reliability, and accessibility across all system components.

Company Logo

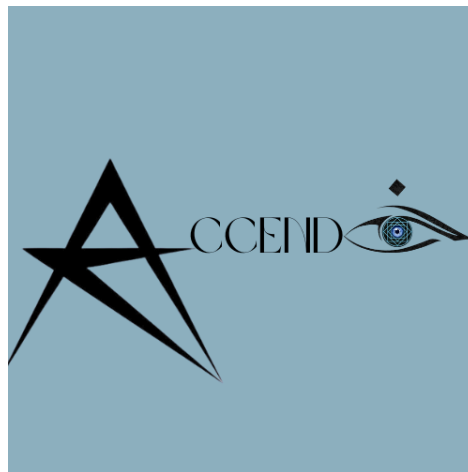


Figure 1. *Official Logo of Accendo Company*

The Accendo logo tells a story of light, vision, and purpose. It embodies the company's mission to illuminate the path of learning for visually impaired individuals through the power of innovation and technology. The letter A, formed as a radiant star, captures the spirit of enlightenment and guidance. It symbolizes how Accendo strives to spark curiosity, inspire confidence, and lead learners toward greater understanding.

At the opposite end, the letter O, creatively shaped as an eye with a diamond above it, reflects clarity, awareness, and the brilliance of knowledge. The eye represents perception



and insight, while the diamond above it signifies the precious value of learning and the pursuit of intellectual growth. Together, these elements form a visual harmony that expresses the company's belief that knowledge should shine for everyone, regardless of limitation.

Through its elegant typography and thoughtful symbolism, the Accendo logo stands as a beacon of inclusivity and empowerment. It reflects the company's vision of transforming education into an experience guided by light, strengthened by understanding, and enriched by accessibility.

Company Location

Accendo is located in Ligid Tipas, Sapang Viejo, Taguig City, Metro Manila, Philippines, a community that fosters growth and innovation. Its strategic location provides accessibility to nearby institutions and potential partners that support technological development and inclusive education initiatives.



Figure 2. Location Sketch of Accendo Company situated in Ligid Tipas, Sapang Viejo, Taguig City, Metro Manila, Philippines.

Product/Service Description

Accendo is a technology-driven company dedicated to designing and developing accessible and intelligent learning platforms for visually impaired learners. It focuses on



bridging the gap between innovation and inclusivity by providing adaptive educational tools that empower learners to achieve academic growth without the barriers of visual limitations.

Through the integration of Artificial Intelligence (AI), Natural Language Processing (NLP), and Multi-Modal Haptic Feedback, Accendo transforms traditional e-learning into a more inclusive, interactive, and human-centered experience. These advanced technologies work together to create a system that not only supports accessibility but also personalizes the learning journey of every user.

The company's flagship product, the Adaptive Learning Management System (LMS) for the Visually Impaired, introduces a revolutionary approach to digital education. It provides a fully interactive environment that caters to learners with visual impairments through innovative features such as:

- Voice-based interactions that allow seamless and hands-free navigation of learning modules.
- Text-to-speech and speech recognition technologies that enable natural and effective communication between the learner and the system.
- Haptic feedback integration that delivers tactile learning cues, allowing users to engage through the sense of touch.
- AI-driven content adaptation that adjusts lessons, assessments, and feedback based on each learner's individual pace and progress.

By combining technology and empathy, Accendo envisions a world where no learner is left behind, regardless of their physical abilities. The Adaptive LMS stands as both a technological product and a service platform, representing Accendo's commitment to inclusive innovation and its mission to make education truly accessible for all.



CHAPTER II

MARKETING ASPECT

The nature of the unsatisfied demand for accessible digital education among visually impaired learners. It analyzes the market conditions, demand-supply situation, research techniques, target market, competition, and marketing strategies essential for the successful launch and sustainability of Accendo's Adaptive Learning Management System (LMS).

Industry Background

The proposed project operates within the Educational Technology (EdTech) sector, focusing on inclusive digital learning solutions for persons with disabilities. Over the past five years, the EdTech industry has grown rapidly due to:

- The rise of online learning platforms (2020–2024)
- Increased government support for digital education and accessibility programs
- Higher demand for assistive learning tools for students with disabilities
- Integration of AI technologies like speech recognition and natural language processing

Despite significant growth in mainstream e-learning, accessibility tools remain underdeveloped, leaving visually impaired learners dependent on traditional assistive devices. This gap creates an opportunity for AI-driven, adaptive e-learning systems that can deliver personalized and accessible learning experiences. Accendo positions itself within the Service and Technology Development sector, supplying an inclusive LMS designed specifically for visually impaired learners.

Market Research

Market research was conducted to determine the demand for an accessible Adaptive LMS and to validate the existence of an unsatisfied need in digital learning accessibility.

1. Research Technique Used



a. Slovin's Formula

Slovin's Formula is used to determine the appropriate sample size from a given population when the population behavior is unknown.

$$n = \frac{N}{1 + Ne^2}$$

where:

- N = total population
- n = sample size
- e = margin of error (commonly 0.05 or 0.10)

This formula allows researchers to generalize results with acceptable accuracy.

b. Probability Sampling Technique

A Stratified Random Sampling method may be used to ensure that visually impaired learners, educators, and institutions are all represented.

Examples of probability sampling:

- **Simple Random Sampling** – everyone has equal chance.
- **Systematic Sampling** – selecting participants at intervals (e.g., every 5th student).
- **Cluster Sampling** – entire groups (schools or organizations) are selected.

2. Other Market Research Sources

- Consultations with accessibility experts
- Interviews with visually impaired students and teachers
- Review of DepEd, CHED, and PDAO (Persons with Disability Affairs Office) data
- Published statistics on assistive learning devices in the Philippines

This mixed-method approach ensures reliable and credible market insights.

Demand Analysis

1. Identification of the Market

The primary and secondary buyers for Accendo's Adaptive LMS are:

Primary Market



- Visually impaired learners enrolled in secondary and tertiary education. These users need accessible digital tools to participate in online learning and perform academic tasks independently. Existing LMS platforms are not fully optimized for non-visual navigation, creating a strong demand for a system designed specifically with accessibility in mind.

Secondary Market

- Educational institutions (public and private schools, universities, and special education centers). Schools are responsible for ensuring compliance with inclusive education policies and are willing to adopt systems that support learners with disabilities.

Tertiary Market

- Government agencies and non-profit organizations, such as:
 - DepEd Inclusive Education Programs
 - CHED Office of Student Development
 - PDAO (Persons with Disability Affairs Office)
 - NGOs supporting visually impaired individuals

These stakeholders keep accessibility as part of their mandate and often fund inclusive learning programs.

2. Potential Demand for the Product

Based on the growing emphasis on digital learning and accessibility programs in the Philippines, the potential demand for Accendo's LMS is high for the following reasons:

a. Rising number of visually impaired learners

Government data reports an increasing enrollment of visually impaired students in mainstream schools due to strengthened inclusive education policies. These learners require specialized digital tools to access online content effectively.

b. Lack of locally developed accessible LMS solutions

Existing LMS platforms (e.g., Canvas, Moodle, Google Classroom) offer basic accessibility features but do not support fully voice-controlled or AI-adaptive interactions. This leaves a large demand gap that Accendo can fill.

c. Schools are seeking compliant accessibility tools



The Department of Education and CHED encourage institutions to adopt accessible technologies to support PWD learners. Schools are actively looking for affordable systems that meet accessibility standards.

d. Increased familiarity with e-learning

Since 2020, online learning has normalized. This increases the demand for tools that enhance digital learning for students with disabilities.

3. Willingness to Pay

Primary Users (Students)

Most visually impaired students rely on free or institution-provided tools. This makes a school-sponsored subscription model preferable. Students benefit at no direct cost.

Secondary Users (Schools/Universities)

Schools demonstrate willingness to pay for:

- Accessibility tools
- Specialized LMS upgrades
- Assistive technologies

Comparable foreign software solutions such as JAWS, screen readers, and accessibility devices cost ₱50,000–₱60,000, making them expensive for Filipino schools. Accendo offers a more affordable, locally developed, subscription-based solution—making schools highly likely to adopt it.

Tertiary Users (Government/NGOs)

Government agencies regularly allocate budgets for:

- Assistive learning devices
- Educational accessibility projects
- ICT modernization programs

Their willingness to fund the LMS is significantly high due to alignment with PWD inclusion policies.

4. Demand Level for the Proposed Product



The estimated demand level for the product is high due to:

- The substantial number of visually impaired learners in need of accessible e-learning tools
- The lack of competing adaptive LMS platforms in the Philippines
- Increasing government support for inclusive education
- The rising acceptance of AI-powered learning solutions

If 10–20% of institutions with visually impaired learners adopt Accendo within the first few years, this already represents a strong and stable demand base.

5. Opportunities

- Untapped market for adaptive LMS specifically designed for visually impaired students
- Low competition from local developers
- Cost advantage compared to imported assistive technologies
- Government support for PWD learning tools
- Potential for expansion to other disabilities and Southeast Asian markets
- Positive social impact, making the system attractive for grant funding and institutional support

6. Threats

- Entry of international tech companies offering similar accessibility enhancements
- School budget limitations that may delay procurement
- Rapid technological advancements requiring continuous updates
- Internet constraints in some Philippine regions affecting system performance
- Resistance to change from institutions used to traditional LMS platforms

Market Size

Accendo targets the following market segments:

Primary Market

1. Visually Impaired Learners (High school, Senior High, College)
 - Demographic: Individuals with complete or partial visual impairment
 - Psychographic: Learners who prefer accessible and adaptive tools
 - Behavioral: Regular users of screen readers, audio modules, voice commands

Secondary Market



2. Educational Institutions

- Schools with Inclusive Education Programs
- Universities offering Accessibility Resource Centers
- Organizations supporting PWD education

Tertiary Market

3. Government Offices & NGOs

- DepEd & CHED
- PDAO
- Foundations supporting blind and visually impaired students

These segments represent thousands of potential Filipino learners who require accessible educational platforms to participate fully in digital learning.

Market Demand

Over the past five years, demand for assistive educational tools has steadily increased due to:

- Shift to e-learning platforms
- Increasing number of visually impaired learners enrolled in mainstream education
- Lack of advanced, AI-driven accessibility systems in the Philippines
- Government support for disability inclusion programs
- Growing awareness of digital accessibility standards

Traditional LMS platforms (e.g., Canvas, Moodle) do not cater specifically to visually impaired learners, creating an unmet demand for specialized solutions.

Accendo offers:

- AI-powered voice navigation
- Speech recognition
- Text-to-speech modules
- Adaptive, personalized content

These features directly respond to the needs of visually impaired learners.



Market Forecast (Next 5 Years)

Assuming continuous growth in digital learning adoption and increasing enrollment of PWD learners, the projected demand for Accendo's system is expected to rise by:

- **Year 1:** Pilot adoption in 3–5 institutions
- **Year 2:** Expansion to 10–15 schools
- **Year 3:** Partnership with government agencies (PDAO, DepEd)
- **Year 4:** Increased adoption due to accessibility mandates
- **Year 5:** National expansion and potential regional (Southeast Asia) use

Projected demand aligns with the market size and annual growth trends of the EdTech sector.

Supply Analysis

Direct Competitors

1. **Screen Reader Software** (JAWS, NVDA)
 - Provides accessibility but no adaptive LMS features
 - Subscription-based
2. **Basic LMS Platforms** (Moodle, Canvas)
 - Offer online classes
 - Limited voice interaction
 - Not designed specifically for visually impaired learners

Indirect Competitors

3. **Assistive Learning Devices**
 - Braille notation devices
 - Audio book apps
 - Talking calculators

These competitors provide accessibility tools but lack integration, AI adaptation, and built-in educational modules tailored for visually impaired students.

Accendo's competitive edge includes:

- AI-powered adaptive content
- Full voice navigation
- NLP-based communication system



- Inclusive learning flow
- More affordable than foreign solutions

Competition Analysis

Competitors offer partial solutions but not a complete accessible LMS specifically designed for visually impaired users.

Competitors' Advantages

- Existing reputation (e.g., JAWS)
- Established user base
- Availability of global updates

Accendo's Advantages

- Specialized for visually impaired Filipino learners
- Fully voice-interactive
- Adaptive AI-powered content
- Lower cost and easier deployment for schools
- Localized language options

Gap Analysis

There is a significant gap between demand (high need for accessible learning) and supply (limited dedicated platforms). This gap opens a strong opportunity for Accendo.

Supplier

1. Producers / Suppliers of Existing Products

a. JAWS – Freedom Scientific (United States)

- Product Type: Screen reader software
- Location: Florida, USA
- Distribution Channel:
 - Official website



- Global software distributors
 - Partnerships with disability organizations
- Target Segment:
 - Visually impaired individuals, academic institutions
- USP:
 - Industry-leading accuracy in speech reading and compatibility with most applications

b. NVDA – NV Access (Australia)

- Product Type: Open-source screen reader
- Location: Queensland, Australia
- Distribution Channel:
 - Free download through official website
- Target Segment:
 - Low-income visually impaired users, schools looking for free tools
- USP:
 - Completely free, community-supported, lightweight

c. Moodle & Canvas (Global LMS Providers)

- Product Type: Learning Management Systems with basic accessibility features
- Location:
 - Moodle: Australia
 - Canvas: United States
- Distribution Channel:
 - Online Download
 - Institutional Licensing
 - Cloud hosting partners
- Target Segment:
 - Schools, Universities, Online academies
- USP:
 - Widely used with strong LMS features, but built for general users—not visually impaired learners

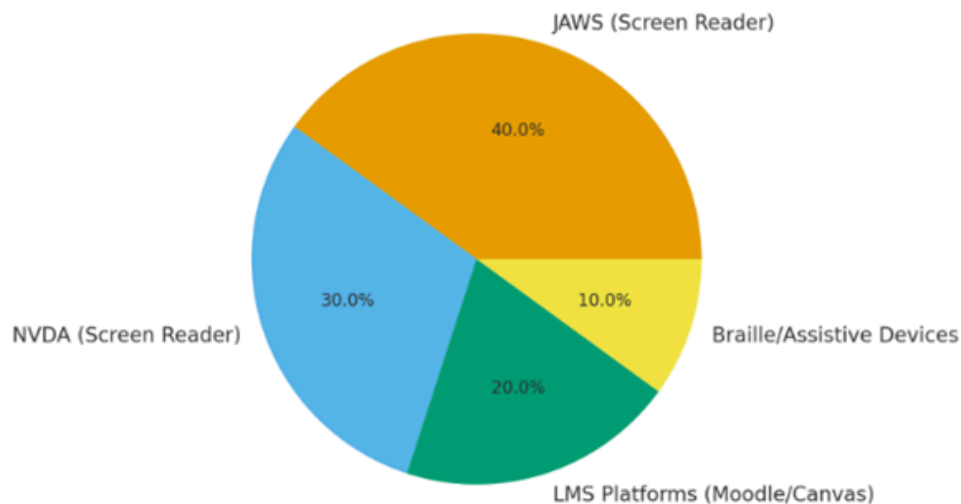
d. Braille Devices & Assistive Hardware (Japan, U.S., Europe)

- Product Type: Braille displays, note-takers, talking devices
- Location:
 - HumanWare (Canada)
 - BrailleSense (Korea)
 - Orbit Reader (U.S.)



- Distribution Channel:
 - Assistive technology stores
 - Online sellers
 - NGO and foundation partnerships
- Target Segment:
 - Students needing tactile reading tools
- USP:
 - Durable hardware for tactile and non-visual learning

Estimated Market Share of Accessibility Tools for Visually Impaired Learners



However, none of these competitors provide:

- A Philippine-developed LMS
- A system designed specifically for visually impaired Filipino learners
- AI-powered adaptive navigation
- Full voice control
- Local pricing

This becomes Accendo's strongest market advantage.

Market Supply

The market supply refers to the total availability of assistive learning tools and accessibility technologies provided to visually impaired learners over the past five



years. These tools include screen readers, basic accessible LMS platforms, and imported assistive hardware such as Braille displays. Since the Philippines does not have a local producer of accessibility-focused Learning Management Systems, the national supply is heavily dependent on foreign suppliers. The annual supply therefore reflects the distribution volume and availability of these existing products.

1. Annual Volume of Supply (Last Five Years)

Over the past five years, the supply of technologies used by visually impaired learners has shown slow but steady growth. The supply mostly comes from foreign companies that distribute their software or hardware globally, including the Philippines.

Existing Supply Sources

1. Screen Readers

- **JAWS** (paid software)
- **NVDA** (free and open-source)

These are the most commonly used tools by visually impaired learners.

2. General LMS Platforms with Limited Accessibility Features

- Moodle
- Canvas

These platforms can be used by visually impaired learners only with the help of screen readers.

3. Assistive Hardware Devices

- Braille displays
- Braille note-takers

These are imported and distributed through specialized organizations.

Estimated Annual Supply Trend

Across these categories, the estimated total annual supply in the Philippines was approximately:

- **2020:** 4,400 units
- **2021:** 5,020 units
- **2022:** 5,890 units
- **2023:** 6,580 units
- **2024:** 7,180 units

These figures represent the combined distribution of software licenses, free downloads, institutional LMS adoptions, and imported assistive devices. Although



the supply increases slightly every year, the growth remains small in relation to the actual national demand.

2. Availability of Supply and Sufficiency

Despite the presence of global suppliers, the supply available in the Philippines is **insufficient**. Several reasons contribute to this shortage:

a. High Dependence on Imported Products

Most tools are foreign-made, making them expensive and difficult for schools to procure.

b. Limited Accessibility Features in Existing LMS Platforms

Moodle and Canvas are not designed specifically for visually impaired learners. Their accessibility depends on third-party screen readers, which places additional burden on institutions.

c. High Cost of Assistive Hardware

Braille devices and tactile learning tools are extremely costly, reducing accessibility for low-income learners.

d. Software Licensing Barriers

Paid screen readers such as JAWS are not affordable for most Filipino students; schools rarely buy them in bulk.

These limitations restrict the availability of accessible learning solutions nationwide.

3. GAP Analysis (Demand vs Supply)

To determine whether the supply is adequate, it must be compared with the estimated needs of visually impaired learners.

Estimated Demand

Based on national disability statistics, there are approximately **100,000 visually impaired learners** in the Philippines, and at least **60,000** require assistive learning technologies to access digital education.

Supply Gap



Even at the highest supply level (7,180 units in 2024), the total available tools fall far short of the estimated demand.

Supply Gap (2024):

60,000 demand – 7,180 supply = 52,820 unmet need

This substantial gap shows that supply covers **less than 15%** of national needs.

4. Effect of the Supply Gap on the Proposed Business

The shortage of accessible digital learning solutions creates a strong opportunity for Accendo's Adaptive LMS:

a. High Demand for a Local, Affordable Solution

Because the current supply is imported and expensive, there is a market opening for a more cost-effective Filipino-developed LMS.

b. Lack of Accessibility-Focused LMS Options

No existing LMS provides built-in voice navigation, adaptive content, or NLP features designed specifically for visually impaired Filipino learners.

c. Strong Institutional Interest

Schools and government agencies have mandates for inclusive education but lack the tools to fulfill them.

Accendo fills this gap.

d. Low Competition

The supply shortage shows that Accendo will face minimal direct competition, allowing faster adoption.

Price

The proposed institutional subscription price of **₱75,000 per year** is strategically positioned to remain affordable while still competitive in the assistive technology and e-learning market. This pricing is justified through the following comparisons:



- **Lower than most foreign LMS tools**, which typically cost **₱100,000 or more per year** for institutional use.
- **Significantly more affordable than JAWS**, which costs **₱50,000–₱60,000 per student (one-time)**—making it unrealistic for large-scale use in Philippine schools.
- **More powerful, specialized, and adaptive than NVDA**, which is free but provides **limited accessibility features** and lacks LMS-level functionality.
- **Attractive for both public and private institutions**, considering its inclusive design and comprehensive accessibility tools tailored for visually impaired learners.

Through this balanced pricing approach, Accendo positions itself as an accessible, cost-effective, and high-value solution for Philippine education institutions.

Institutional Subscription (Primary Pricing Model)

₱75,000 per Year

Applicable for:

- Schools
- Universities
- Learning centers
- SPED institutions

Inclusions

- Full access to the Accendo Adaptive LMS
- NLP-powered voice navigation
- Teacher and Administrator dashboards



- Unlimited student accounts within the institution
- Dedicated technical support
- Cloud hosting and system maintenance
- Continuous accessibility and security updates

This subscription structure ensures that institutions receive complete functionality without hidden or additional mandatory fees.

Profitability Model

Accendo's profitability is driven by **recurring subscription income**, **low operational costs**, and **scalable technology**. As a cloud-hosted LMS, the system allows the business to support multiple schools without significantly increasing overhead.

1. Recurring Revenue Through Subscriptions

The core business model is subscription-based, ensuring predictable and consistent yearly income.

Projected Annual Revenue Based on Acquired Institutions:

- **10 institutions** × ₱75,000 = **₱750,000 per year**
- **20 institutions** × ₱75,000 = **₱1,500,000 per year**
- **50 institutions** × ₱75,000 = **₱3,750,000 per year**

Because **student access is free**, schools are more likely to adopt the platform, accelerating user growth and market expansion.

SWOT Analysis



Strengths

- AI-powered NLP system
- Voice-based navigation
- Locally developed, affordable
- Designed specifically for visually impaired learners

Weaknesses

- New brand in the EdTech industry
- Requires internet for some features
- Needs continuous AI updates
- Limited initial market reach

Opportunities

- Demand for inclusive education
- Government support for PWD programs
- Expansion to Southeast Asia
- Rising awareness on accessibility

Threats

- Entry of global accessibility tools
- Technological advancements from competitors
- Budget limitations in schools
- Rapid EdTech competition

Marketing Strategy

Accendo will use the following strategies:



Pricing Strategy

- **Value-based pricing**

Pricing is based on:

- Accessibility improvement
 - Integration of AI tools
 - Customization for schools
- Affordable subscription to encourage institutional adoption.

Partnership and Collaboration Strategy

Accendo recognizes that strong partnerships are essential for establishing credibility and expanding its reach in the accessibility and education sectors. The team will implement a structured partnership strategy focusing on government institutions, disability advocacy groups, SPED schools, and educational organizations.





College of Computer Studies
Alkalde Jose St. Kapasigan Pasig City, Philippines 1600
628-1014 Loc. 106 officeoftheoic@plpasig.edu.ph

PAMANTASAN NG LUNGSOD NG PASIG



**UP COLLEGE OF EDUCATION
SPECIAL EDUCATION AREA**

presents

**INCLUSIVE AND SPECIAL NEEDS EDUCATION
SEMINAR SERIES (PART 1)**

ISNED

**Actualizing Inclusion of
Learners with Disabilities (LWD)
in Philippine General Education Schools**

29 June 2024 (Saturday), 8:00 am - 5:00 pm
UP School of Statistics Auditorium
Diliman, QC

Seminar Fee: Php 1,500

Register here: <https://forms.gle/K23QmuMYtCTtgxuX6>



For inquiries, email: educsped.upd@up.edu.ph



UP SPED Inclusive and Special Needs Education Seminar Series

SCAN ME



Get
CPD
points!



1. Partnership with PWD and Disability Advocacy Organizations

Target Partners

- Philippine Blind Union (PBU)
- Resources for the Blind, Inc. (RBI)
- National Council on Disability Affairs (NCDA)
- Local Persons with Disability Affairs Offices (PDAO)

How the Team Will Execute the Partnership

1. Formal Introduction & Proposal

- Send formal partnership letters outlining Accendo's purpose and the benefits of collaboration.
- Request meetings to introduce the LMS and discuss possible pilot testing.

2. Demonstration Sessions

- Conduct on-site or virtual demos specifically for their members and trainers.
- Allow PWD representatives to test the NLP navigation feature and provide feedback.

3. Co-Hosted Training Workshops

- Organize joint workshops focusing on digital literacy, accessible e-learning, and assistive tools.
- Provide free temporary access to the LMS for training purposes.

4. Gather Testimonials & Impact Reports



- Collect feedback and testimonials from visually impaired testers.
- Use these for marketing, credibility building, and validating the system's effectiveness.

2. Collaboration with Educational Institutions (SPED, Colleges, and Universities)

Target Partners

- SPED centers
- Public and private universities
- State colleges
- ICT and education departments

How the Team Will Execute the Partnership

1. Campus Outreach

- Contact SPED coordinators and school administrators.
- Schedule on-campus demonstrations where faculty and students can test the LMS.

2. Pilot Program Implementation

- Offer a free 1–2 month pilot access to selected schools.
- Provide training manuals and a system walkthrough for teachers.

3. Faculty Training

- Conduct workshops for teachers and administrators on how to integrate the LMS into their classes.



- Provide facilitator kits, video tutorials, and live support.

4. Evaluation & Conversion

- After the pilot period, gather feedback, present improvements, and encourage subscription.
 - Offer discounted introductory plans to early-adopter schools.
-

3. Partnership with DepEd and CHED

How the Team Will Execute the Partnership

1. Submission for Accreditation

- Prepare a technical and pedagogical document presenting Accendo as an assistive learning tool.
- Submit this to DepEd ICTS and CHED's Office of Programs and Standards for evaluation.

2. Participation in Government Programs

- Apply to join DepEd's digital transformation programs.
- Attend CHED education fairs, seminars, and conferences as exhibitors.

3. Regional Rollout Strategy

- Collaborate with DepEd regional offices to deploy the LMS to public schools.
- Provide free trials to selected public SPED institutions to showcase impact.

4. Endorsement and Scaling



- Aim to secure endorsements allowing Accendo to be recommended across divisions.
- Use the recommendation to scale to more public schools nationwide.

4. Collaboration with Local Government Units (LGUs)

How the Team Will Execute the Partnership

1. Coordination with LGU Education & PWD Offices

- Present Accendo to mayors, education committees, and PWD officers.
- Showcase how it supports inclusive education within the city/municipality.

2. LGU-Funded Pilots

- Offer agreements where LGUs fund the LMS subscription for public schools in their area.
- Propose packages that support multiple institutions within the locality.

3. Community Outreach Programs

- Work with LGUs for community-based digital literacy training.
- Conduct free sessions for visually impaired residents and students.

5. Strategic Partnerships with NGOs and Private Foundations

How the Team Will Execute the Partnership

1. Identify CSR Programs



- Target foundations with education, accessibility, or digital inclusion CSR initiatives.

2. Funding Collaborations

- Propose that the foundation sponsor LMS access for a partner school or community.

3. Joint Events

- Conduct events or webinars funded by partner organizations to expand reach.

Summary of How the Team Will Implement Partnerships

- Write and send official proposals
- Conduct on-site and virtual demos
- Offer pilot access for evaluation
- Run training workshops
- Use feedback for product refinement
- Convert partners into long-term subscribers
- Expand collaborations through government and NGOs

4Ps of Marketing

1. Product

- Adaptive Learning Management System
- AI-powered NLP



College of Computer Studies

Alkalde Jose St. Kapasigan Pasig City, Philippines 1600
628-1014 Loc. 106 officeoftheoic@plpasig.edu.ph

PAMANTASAN NG LUNGSOD NG PASIG

- Speech recognition, text-to-speech
- Accessible modules for visually impaired learners

2. Place

- Distributed through:
 - Schools and universities
 - Government accessibility programs
 - Online platform
 - NGOs and foundations supporting the blind

3. Price

- Affordable subscription model
- Lower than foreign assistive devices
- Free student user accounts via school partnership

4. Promotion

- Social media promotions
- Free demos
- Institutional presentations
- Accessibility webinars
- Collaboration with blindness support groups



College of Computer Studies

Alkalde Jose St. Kapasigan Pasig City, Philippines 1600
628-1014 Loc. 106 officeoftheoic@plpasig.edu.ph

PAMANTASAN NG LUNGSOD NG PASIG

CHAPTER III TECHNICAL ASPECT

This chapter presents the technical requirements, production process, facilities, and resources necessary for the development and delivery of Accendo's Adaptive Learning Management System (LMS) for visually impaired learners. It also explains the technology involved, materials needed, production flow, project layout, and waste management programs to ensure sustainable system operations.

Production / Service Process

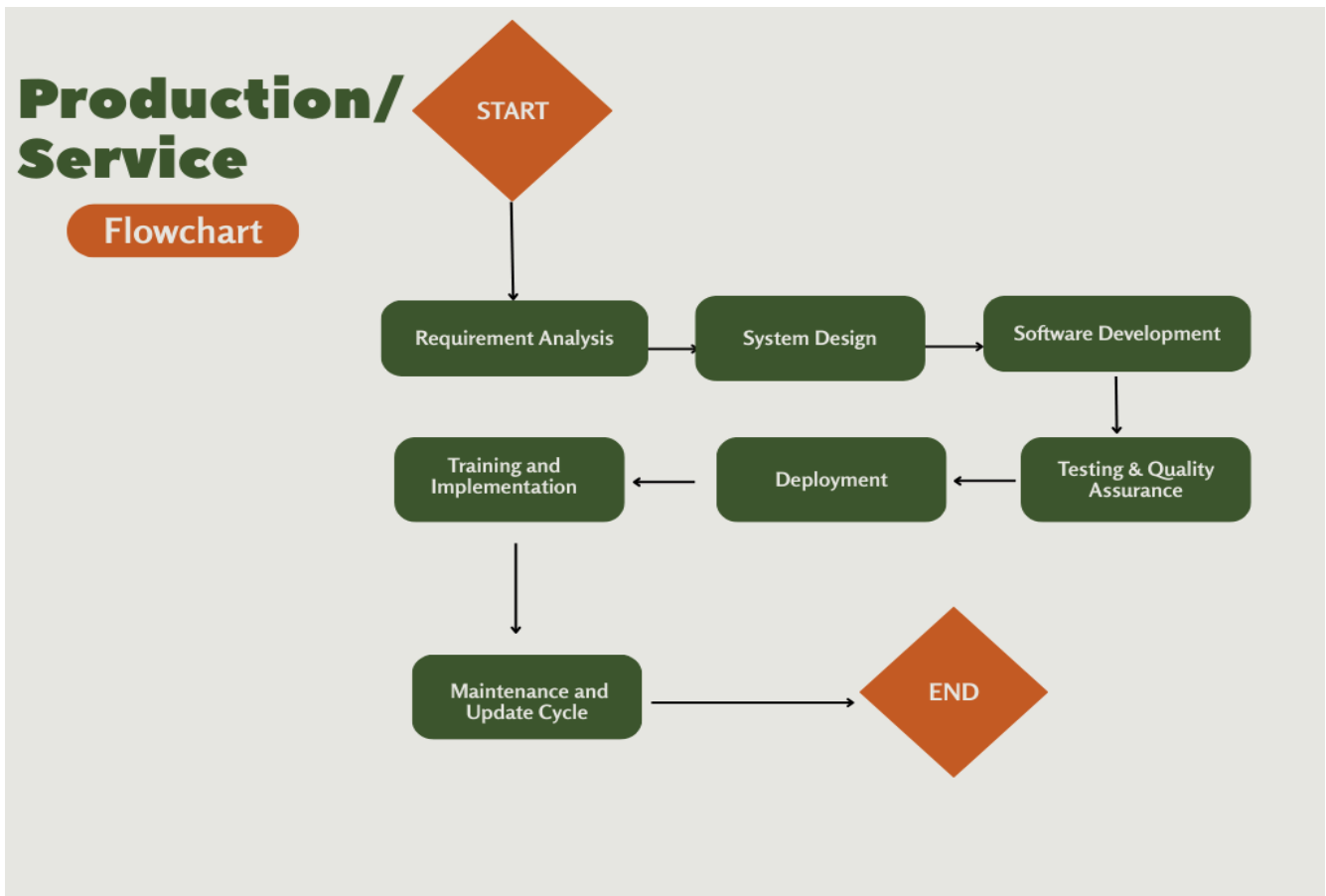
Accendo provides a technology-based service, specifically the development and deployment of an AI-powered Adaptive LMS. As a digital product, the production process involves software development, integration of assistive technologies, testing, deployment, and continuous system maintenance.

Below is the step-by-step production process:



1. Requirement Analysis

- Identify needs of visually impaired learners



- Gather requirements from teachers, schools, and accessibility experts
- Review accessibility standards (WCAG, ISO/IEC 25010)

2. System Design

- Create UI/UX based on voice-based interactions
- Design AI-NLP modules
- Draft system architecture (database, backend, frontend)



3. Software Development

- Develop voice recognition features
- Integrate text-to-speech and AI adaptive content
- Code LMS modules such as:
 - Course creation
 - Assessment tools
 - Voice navigation system
 - Personalized learning adaptation

4. Testing and Quality Assurance

- Conduct functional testing
- AI performance testing
- Accessibility testing with visually impaired beta users
- Debugging and system optimization

5. Deployment

- Install system on cloud servers
- Configure institution accounts
- Create admin, teacher, and student portals
- Install necessary accessibility packages

6. Training and Implementation



- Conduct training sessions for teachers
- Orientation for visually impaired students
- Provide user manuals and voice-guided tutorials

7. Maintenance and Update Cycle

- System monitoring
- Regular updates for AI and NLP models
- Continuous improvement based on feedback

Production / Service Flowchart

Production / Service Facilities

Although Accendo is a digital service provider, it still requires a functional workspace that includes development equipment and operational areas.

A. Office & Workspace Components

1. Development Area

- Computers for programmers
- Testing devices (laptops, mobile phones, tablets)
- Whiteboards and planning boards

2. Server & Data Management Area

- Cloud server management consoles
- Backup storage
- Secure network infrastructure



College of Computer Studies
Alkalde Jose St. Kapasigan Pasig City, Philippines 1600
628-1014 Loc. 106 officeoftheoic@plpasig.edu.ph

PAMANTASAN NG LUNGSOD NG PASIG

3. Training Room

- Used for system demonstrations
- Orientation for teachers and visually impaired learners
- Equipped with audio setups and screen reader tools

4. Administrative Area

- Workplace for administration, finance, and documentation
- Manager and HR desks

5. Meeting/Collab Room

- Brainstorming sessions
- Client meetings
- Agile sprint planning

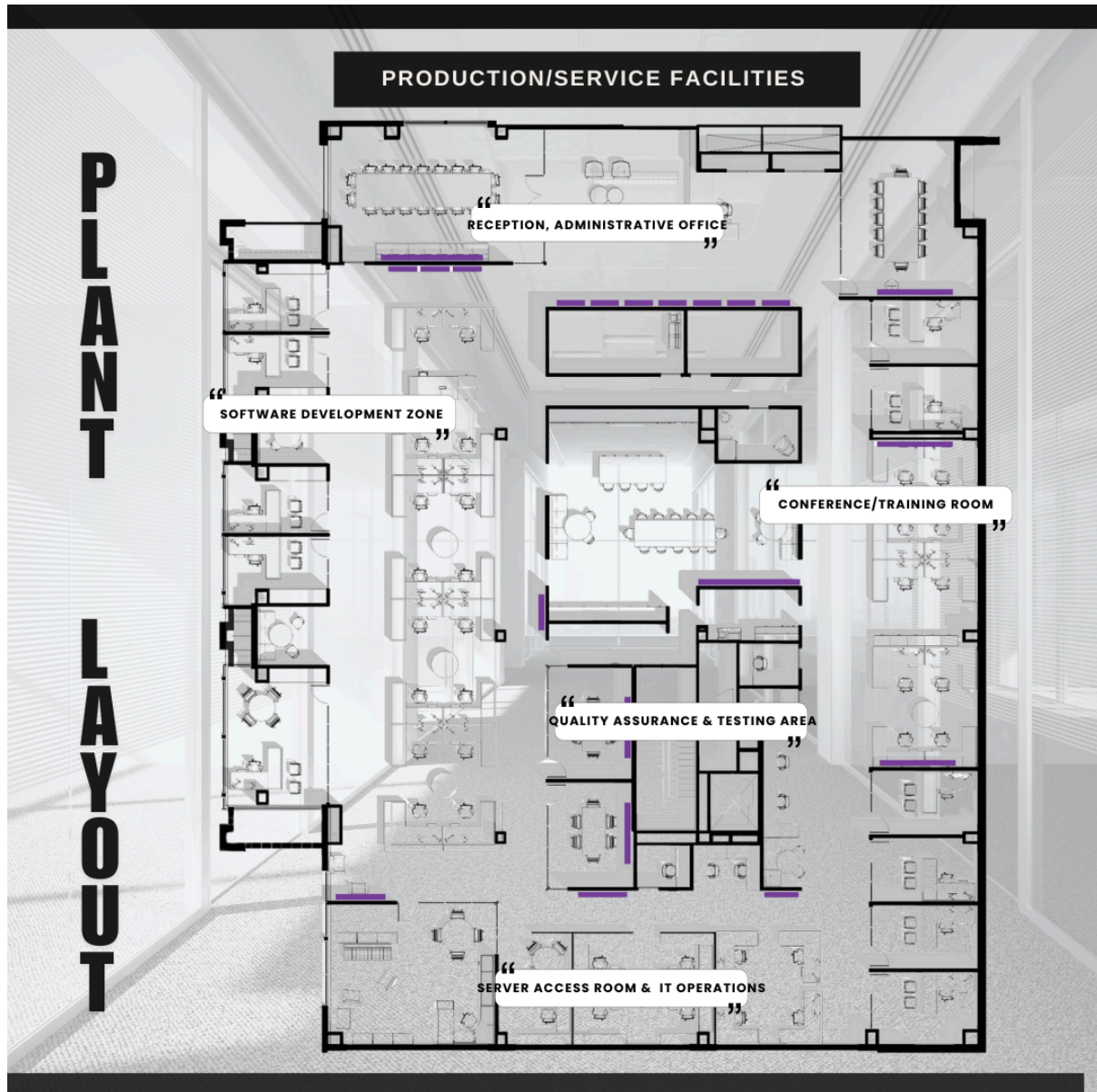


College of Computer Studies

Alkalde Jose St. Kapasigan Pasig City, Philippines 1600
628-1014 Loc. 106 officeoftheoic@plpasig.edu.ph

PAMANTASAN NG LUNGSOD NG PASIG

B. Plant Layout



The floor layout is designed to maximize productivity and support the software development life cycle.

- **Front Section:** Reception, Administrative Office
- **Left Wing:** Software Development Zone
- **Right Wing:** Quality Assurance & Testing Area
- **Back Section:** Server Access Room and IT Operations



- **Conference/Training Room:** Near the entrance for visiting clients and trainees

This arrangement minimizes workflow delays and improves communication among departments.

SOURCES AND COST RAW MATERIALS

A. Raw Material List and Sources

Material / Resource	Source	Estimated Cost
Cloud Hosting (AWS / Google)	International providers	₱8,000 – ₱15,000/month
Development Laptops	Local computer shops	₱30,000 – ₱45,000/unit
AI & NLP Libraries (open-source)	Github, Open-source	Free
Speech-to-Text API	Google Cloud, Azure	₱3,000 – ₱7,000/month
Text-to-Speech Engine	Google/Azure	₱2,000 – ₱5,000/month
Software Tools (VSCode, Figma, Postman)	Online	Mostly Free



Database Hosting	Cloud providers	₱1,500 – ₱5,000/month
Office Utilities	Local suppliers	₱5,000 – ₱10,000/month
Training Materials & Manuals	Internal production	₱2,500 – ₱5,000

B. Operational Expenses

Category	Estimated Monthly Cost
Internet	₱2,000 – ₱3,500
Electricity	₱1,500 – ₱2,500
System Maintenance	₱3,000 – ₱8,000
Staff Salaries	Based on team composition
Software Maintenance & API Fees	₱8,000 – ₱15,000



Waste Disposal Program

Although Accendo is a digital business, it still generates minimal waste such as electronic waste (e-waste), paper waste, and unused digital files. To ensure environmental responsibility, the company implements the following waste disposal programs:

A. Digital Waste Management

- Regular deletion of redundant data
- Cloud storage optimization
- Secure disposal of outdated files
- Encryption of sensitive information
- Data cleanup every development cycle

B. Physical Waste Management

- Recycling of paper and ink cartridges
- Proper labeling and segregation of trash
- Reduction of printed documents (paperless office policy)

C. Electronic Waste (E-Waste) Disposal

- Donate functional older devices to schools or NGOs
- Partner with certified e-waste recycling companies
- Proper disposal of damaged hardware components

D. Energy Efficiency Program

- Use of low-power devices
- Scheduled shutdown of development computers
- Use of LED lighting and energy-efficient equipment



CHAPTER IV

FINANCIAL ASPECT

This chapter discusses the overall financial position of Accendo in terms of projected expenses, funding requirements, operational costs, profitability, and financial stability. It presents the financial assumptions, financial statements (Cash Flow Statement, Income Statement, and Balance Sheet), and an analysis of the company's liquidity, efficiency, and long-term viability.

Accendo's financial projection is based on the cost of developing and maintaining the AI-powered Adaptive Learning Management System (LMS), its annual subscription revenue, and the operational expenses required to sustain the business

Introduction

The Financial Aspect provides a comprehensive analysis of Accendo's projected financial performance, funding requirements, and long-term sustainability as it develops and operates its AI-powered Adaptive Learning Management System (LMS) for visually impaired learners. This chapter evaluates how the company will allocate resources, manage expenses, and generate revenue through its subscription-based business model aimed at educational institutions, SPED centers, and government agencies.

In alignment with the technical and marketing strategies discussed in previous chapters, this section outlines the financial assumptions used to build Accendo's financial projections, including development costs, operational expenses, subscription income, and initial capital requirements. It also presents the core financial statements—Cash Flow Statement, Income Statement, and Balance Sheet—to assess the viability and profitability of the business during its initial operating year and beyond.

Furthermore, the chapter includes analyses of the company's liquidity, solvency, and operational efficiency, providing a clear picture of Accendo's ability to meet financial obligations, sustain operations, and scale its services over time. Through these evaluations, the Financial Aspect demonstrates that Accendo is positioned to operate sustainably, maintain strong financial health, and support ongoing improvements to its accessible learning platform as more institutions adopt the system.



Financial Assumptions

The following assumptions guide the creation of the financial statements:

1. Revenue Model

- Accendo charges **₱75,000 per year per institution** for unlimited student accounts.
- Students do not pay any fees; institutions are the paying clients.

2. Sales Projection (First Operational Year)

- Conservative estimate: **3 institutions** in Year 1
- Moderate estimate: **5 institutions**
- Full target estimate: **10 institutions**

3. Development and Operational Costs

Based on Chapter 3 technical requirements:

- Cloud hosting: ₱8,000–₱15,000 per month
- Speech-to-text & text-to-speech APIs: ₱5,000–₱12,000 per month
- Office utilities: ₱5,000–₱10,000 per month
- Internet & electricity: ₱3,500–₱6,000 per month
- Developer and staff salaries vary by team size

4. Initial Capital

Accendo is assumed to begin with **₱300,000–₱500,000 seed capital** to cover development equipment, cloud hosting, and early operations.

5. Operating Expenses Increase

A 5% yearly increase is assumed for inflation and technology upgrades.

Financial Statements



A. Cash Flow Statement

Shows how money enters and leaves the business.

Projected Cash Inflows (Year 1)

Source	Amount
Subscription Revenue (3–10 institutions)	₱225,000 – ₱750,000
Initial Capital	₱300,000 – ₱500,000
Total Cash Inflows	₱525,000 – ₱1,250,000

Projected Cash Outflows (Year 1)

Expense Category	Estimated Annual Cost
Cloud Hosting	₱96,000 – ₱180,000
Speech/Text API Fees	₱60,000 – ₱144,000
Database Hosting	₱18,000 – ₱60,000
Office Utilities	₱60,000 – ₱120,000
Internet & Electricity	₱42,000 – ₱78,000
System Maintenance	₱36,000 – ₱96,000



Staff Salaries (small team estimate)	₱300,000 – ₱600,000
Total Cash Outflows	₱612,000 – ₱1,278,000

Cash Flow Summary (Year 1)

- Low adoption scenario: –₱87,000 (near break-even)
- Moderate adoption: +₱138,000
- High adoption: +₱638,000

B. Income Statement

Shows revenues, expenses, and net income.

Projected Income Statement (Year 1)

Category	Amount
Revenue (₱75,000 × 5 institutions)	₱375,000
Less: Operating Expenses	₱300,000 – ₱500,000
Net Income (Annual)	₱75,000 – ₱175,000



C. Balance Sheet

ACCENDO Corporate Finance Institute®. All rights reserved.					
Balance Sheet					
[PHP \$ millions]					
	2026	2027	2028	2029	2030
Assets					
Current assets:					
Cash on Hand & Bank	170,000	210,000	255,000	300,000	350,000
Development Equipment	95,000	100,000	105,000	110,000	115,000
Software Assets (LMS)	200,000	190,000	180,000	170,000	160,000
Inventory	8,000	8,500	9,000	9,500	10,000
Total current assets	473,000	508,500	549,000	589,500	635,000
Property & Equipment	45,500	43,000	40,600	38,300	36,000
Total Assets	518,500	551,500	589,600	627,800	671,000
Liabilities					
Current liabilities:					
Accounts Payable	20,000	22,000	24,000	26,000	28,000
Accrued expenses	8,000	9,000	10,000	11,000	12,000
Unearned revenue	15,000	18,000	20,000	22,000	24,000
Total current liabilities	43,000	49,000	54,000	59,000	64,000
Long-term Obligations	-	-	-	-	-
Total Liabilities	43,000	49,000	54,000	59,000	64,000
Shareholder's Equity					
Equity Capital	110,000	110,000	110,000	110,000	110,000
Retained Earnings	365,500	392,500	425,600	458,800	497,000
Shareholder's Equity	475,500	502,500	535,600	568,800	607,000
Total Liabilities & Shareholder's Equity	518,500	551,500	589,600	627,800	671,000

Accendo ends the year with strong equity due to low liabilities.

Financial Analyses

A. Solvency Analysis

Solvency refers to Accendo's ability to meet long-term financial obligations and continue its operations over an extended period. Based on the company's



projected Balance Sheet, Accendo holds total assets amounting to approximately ₱390,000–₱635,000 at the end of Year 1, while its total liabilities remain relatively low at ₱30,000–₱80,000. This results in a strong equity position ranging from ₱360,000 to ₱555,000, which indicates that the company is not heavily dependent on debt to fund its operations.

Since Accendo operates primarily through cloud-based services and software, it does not require high-cost physical infrastructure. This reduces the need for long-term loans or equipment financing. The subscription-based revenue model also strengthens solvency because annual institutional payments provide predictable cash inflows that can support future development, maintenance, and expansion.

Additionally, because the system is software-based, the risk of asset depreciation is lower compared to hardware-intensive companies. The majority of investments—such as development costs, hosting expenses, and API subscriptions—are operational rather than capital-intensive. These financial characteristics significantly contribute to Accendo's ability to sustain long-term operations even under fluctuating market conditions.

B. Liquidity Analysis

Liquidity measures the company's capacity to cover short-term financial obligations using available cash and near-cash assets. Accendo's cash inflows for the first year range from **₱225,000 to ₱750,000** depending on institutional adoption, supplemented by initial capital investment. Monthly operational expenses—including cloud hosting, utilities, API fees, and maintenance—range from **₱50,000 to ₱80,000**, which the company can support through the subscription model.

Under moderate adoption (5 institutions), the company generates **₱375,000 annually**, translating to approximately **₱31,250 per month**. When combined with initial capital and controlled expense management, this provides sufficient liquidity to sustain operations. Even in the low adoption scenario (3 institutions), Accendo experiences only a small deficit and remains close to break-even because a large portion of its expenses can be scaled depending on system usage.

The company's operational structure also improves liquidity. Cloud hosting and API consumption can be adjusted based on demand, allowing Accendo to avoid overspending during early adoption stages. Since salaries, hosting costs, and utilities represent the largest financial commitments, these can be planned and staggered according to cash availability.



C. Efficiency Analysis

Efficiency refers to how effectively Accendo uses its assets and resources to generate revenue and maintain operations. The company demonstrates high operational efficiency due to its reliance on cloud-based infrastructure and open-source development tools. This setup reduces upfront capital expenditures and enables rapid scaling without proportional increases in cost.

Accendo's LMS is designed to be subscription-based, meaning that once the system is developed and deployed, additional institutions can be onboarded with minimal marginal cost. For example, expanding from 5 to 10 institutions does not require doubling operational expenses because the system runs on scalable cloud resources. This creates a cost-efficient growth model where revenue can increase faster than expenses.

Additionally, the use of AI and NLP tools—many of which are open-source—reduces licensing costs. API fees for speech recognition and text-to-speech are usage-based, which ensures the company pays only for what it needs. This prevents financial waste and encourages efficient resource allocation.

Asset utilization is also optimized. Hardware assets such as development laptops have long useful lives and require minimal replacement. Software assets like code libraries, LMS modules, and system architecture do not depreciate rapidly, making them highly efficient long-term investments.

Finally, Accendo's operational strategy emphasizes partnerships with schools, LGUs, and organizations. These partnerships reduce marketing and acquisition costs, improving financial efficiency by securing clients at lower expenses compared to traditional advertising campaigns.